RPO1 – REFLECTION POSITIVITY TO ROTATIONS: EXPANDED PROOF OUTLINE

Abstract. This outline decomposes the passage from the prescribed coordinate and diagonal reflection families to rotation covariance of Schwinger functions away from coincidences. The principal steps are spectral support, analytic continuation, positivity and propagation from ordered configurations. Each step links to its hypotheses and proof tasks.

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1. Introduction and target statement

This outline follows RPM1 with the growth condition in Glimm–Jaffe's regularity axiom. Source links identify the current theorem and estimates. The scalar application at the end is a separate conditional supplement from the preserved 27-page edition.
RPM1 manuscript; paired TeX. The earlier 27-page edition is retained for the historical scalar application.

A. From the prescribed RP mirrors to OS covariance

Interface. Overview Target.

Let $d \geq 2$ and let $\mathcal{T}$ be a finite-dimensional real orthogonal $O(d)$ representation whose complex planar rotation weights belong to $\{-2,-1,0,1,2\}$ in every coordinate plane. Let μ be a Borel probability law on $\mathcal{S}'(\mathbb{R}^d;\mathcal{T})$ and F its real random field. Use the tensor pullback $(g^*\chi)(z) = \rho(g)\chi(gz)$; ρ is the pullback action on indices, so $\rho(g_1g_2) = \rho(g_2)\rho(g_1)$ and $(g_1g_2)^* = g_2^*g_1^*$.

Target theorem. Suppose that μ is invariant under all translations and, for some $C > 0$,

$$\mathbb{E}_\mu e^{F(\chi)} \leq e^{C(\|\chi\|_1 + \|\chi\|_2^2)} \qquad (\chi \text{ any real Schwartz test}).$$

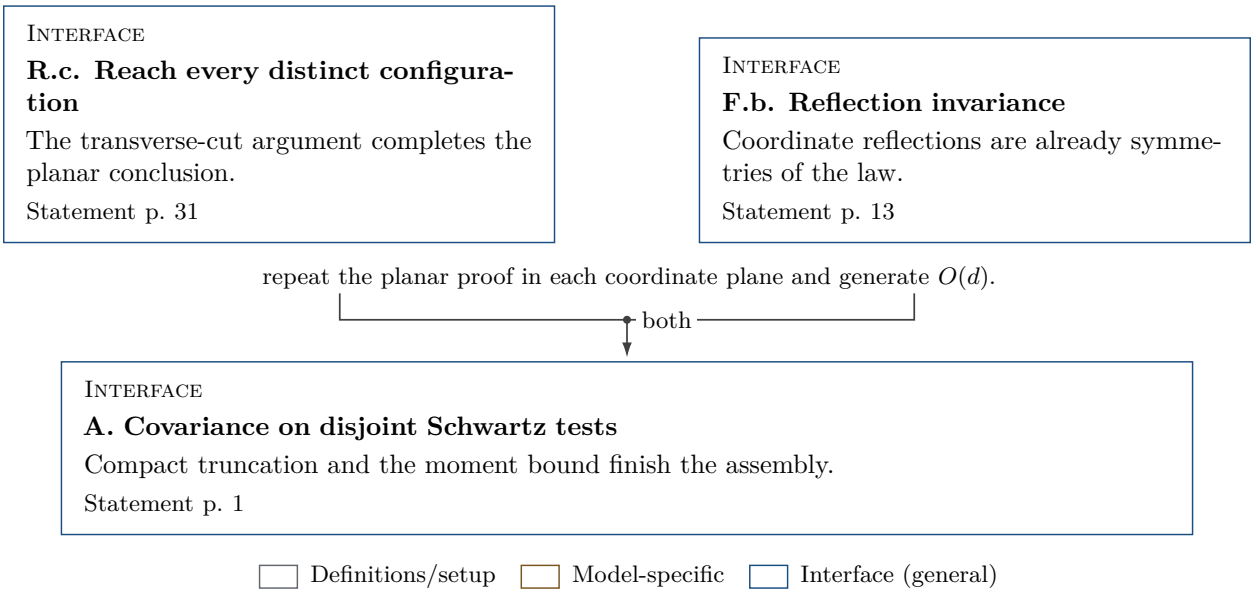
Suppose also that its characteristic functional Φ is reflection positive at each plane

$$x_j = 0 \ (0 \leq j < d), \qquad x_j \pm x_k = 0 \ (0 \leq j < k < d).$$

Precisely, for every finite real test family supported in one open side of such a plane with reflection Θ, $[\Phi(\chi_a - \Theta^*\chi_b)]_{a,b}$ is positive semidefinite. Then, with $S_N(\chi) = \mathbb{E}\prod_j F(\chi_j)$,

$$S_N(g^*\chi_1, \dots, g^*\chi_N) = S_N(\chi_1, \dots, \chi_N) \quad (g \in O(d))$$

whenever the Schwartz tests have pairwise disjoint supports. Equivalently, every covariance defect is supported on the full big diagonal $\Delta = \{z_j = z_k \text{ for some } j \neq k\}$. There are d^2 prescribed planes, hence nine in dimension three.



About 1 page for statement and final assembly. The proof estimates on the map exclude linked inputs.

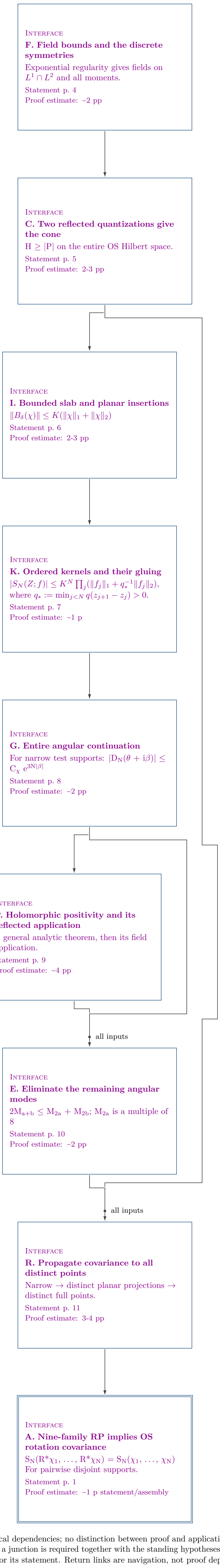
Source. RPM1, Theorem A, (4).

Scope and context. The hypotheses require no field equation or closedness. The conclusion allows noncovariant contact terms. The supplied sharpness examples are in A.scope; the scalar lattice application is in A.application. These pages supply context rather than proof inputs.

RPO1 – Reflection positivity to rotations: expanded proof outline

Definitions/setup Model-specific Interface (general)

Source: RPM1, with the $L^1 + L^2$-squared exponential bound. Full hypotheses and linked source results appear on the statement pages.



Arrows record mathematical dependencies; no distinction between proof and application inputs is made here. Every incoming branch at a junction is required together with the standing hypotheses on the statement page. Click a box for its statement. Return links are navigation, not proof dependencies.

The hypotheses prescribe nine mirror families in dimension three, and d^2 families in dimension d. The planar coefficient weights are at most two.

F. Field bounds and discrete symmetries

Interface. Overview Target.

Two general consequences provide the field domain, moments and discrete symmetries used by the proof. The second needs no exponential hypothesis.

Interface

F.a. Exponential control of fields and moments

Extend fields to $L^1 \cap L^2$ tests and obtain tempered moment kernels.

Statement p. 12

Interface

F.b. Translation invariance and RP imply reflection invariance

Apply this at the prescribed mirrors to obtain quarter-turn symmetry.

Statement p. 13

Definitions/setup Model-specific Interface (general)

These are independent implications. Together with the prescribed coordinate RP they give the positive quotient and dense polynomial classes in C.a. The moment bound is also used to identify and approximate every later partially smeared field expression.

$$\|F(\chi)\|_{L^p(\mu)} \le K(C)(\|\chi\|_1 + \sqrt{p}\,\|\chi\|_2),$$
$$|S_N(\chi)| \le (K(C)\sqrt{N})^N \prod_j (\|\chi_j\|_1 + \|\chi_j\|_2).$$

The reflection result is an invariance statement about the input law. The rotation conclusion in A concerns only disjoint-support moments.

Full write-up estimate: about 2 pages, including shared field conventions.

Source. RPM1, (13), Appendix A, Proposition B.1.

3. Two reflected quantizations give the spectral cone

C. Two reflected quantizations give the spectral cone

Interface. Overview Target.

Take $t = x_0$, $x = x_1$ and $y \in \mathbb{R}^{d-2}$. A stationary field with the field bounds of F.a and RP at $t = 0$ gives one OS Hilbert space. RP at $t + x = 0$ gives a second one for the same ordinary moments.

Interface

C.a. The OS quotient and its translation operators

A contraction semigroup in time and commuting spatial unitaries.

Statement p. 14

Compare the same compact polynomial matrix element in a diagonal frame.

Interface

C.b. A two-measure cone criterion and its field application

Laplace probes give $H \geq |P|$ on the entire OS Hilbert space.

Statement p. 15

Definitions/setup Model-specific Interface (general)

The consequence is a contractive complex translation family on

$$q(z) = \Re z_t - |\Im z_x| \geq 0,$$

norm holomorphic where $q > 0$. It feeds the planar insertion I.b and, on a separate branch, the group-translation analyticity R.a. The latter does not require insertions or angular growth.

Full write-up estimate: 2–3 pages.

Source. RPM1, Section 4, Proposition 4.1.

4. Bounded slab and planar insertions

I. Bounded slab and planar insertions

Interface. Overview Target.

Use the exponential bound of F.a. Work in the t-reflected Hilbert space C.a, with the complex translations of C.b where needed. Below K depends only on the constant C in that bound.

Interface

I.a. A source in a slab gives a bounded field insertion

Reflected source doubling gives an OS norm estimate; differentiate by Cauchy's formula.

Statement p. 16

Localize only the two planar variables.

Interface

I.b. Planar insertion bounded by the adjacent values of q

An L^2 duality and holomorphic mean-value estimate replace the shrinking averaging scale.

Statement p. 17

Definitions/setup Model-specific Interface (general)

For $f \in (L^1 \cap L^2)(\mathbb{R}^{d-2}; \mathcal{T})_{\mathbb{C}}$ and $\xi, \eta \in \mathcal{W}$, the output is a norm-holomorphic bounded operator $A_f(\xi, \eta)$,

$$\|A_f(\xi,\eta)\| \le K\left(\|f\|_1 + \frac{\|f\|_2}{\min(q(\xi), q(\eta))}\right), \qquad q(z) = \Re z_t - |\Im z_x|.$$

The transverse variables remain smeared. The exact smearing and gap-splitting identities are needed to build the kernels in K.a.

Full write-up estimate: 2–3 pages, excluding the OS and spectral inputs.

Source. RPM1, Section 6.2, Section 6.3.

5. Ordered kernels and gluing on verified overlaps

K. Ordered kernels and gluing on verified overlaps

Interface. Overview Target.

With transverse profiles still smeared, the planar insertion yields an actual holomorphic moment kernel on each ordering domain.

Interface

K.a. The ordered kernel and its exact moment identity

Input: I.b, including smearing and inner-gap splitting.

Statement p. 18

Use entry permutations and the quarter turns from F.b.

Interface

K.b. Join the ordering domains

Agreement with real smeared moments fixes the kernels on every overlap.

Statement p. 19

Definitions/setup Model-specific Interface (general)

For $N \geq 2$ and $z_{j+1} - z_j \in \mathcal{W}$ the estimate is

$$|S_N(Z; f)| \leq K^N \prod_j (\|f_j\|_1 + q_*^{-1} \|f_j\|_2), \qquad q_* := \min_{j<N} q(z_{j+1} - z_j).$$

Each factor has an L^1 term and an L^2 term multiplied by q_*^{-1}, independently of the ambient dimension. The joined domain contains the real configurations where either all times or all x coordinates are distinct. The next task uses particular narrow rotation orbits in this domain.

Full write-up estimate: about 1 page with insertion identities available.

Source. RPM1, Section 6.4, Section 6.5.

6. Entire angular continuation on narrow tests

G. Entire angular continuation on narrow tests

Interface. Overview Target.

For clockwise planar rotations $R(\theta)$, write

$$D_N[\chi](\theta) = S_N(R(\theta)^*\chi) - S_N(\chi).$$

A planar configuration is narrow if, in one ordering, its successive displacements lie within angle $\alpha < \pi/4$ of one direction. Use small compact product neighborhoods retaining this property.

Interface

G.a. Cover the entire complex-angle cylinder

Inputs: K's kernels and planar weights at most two. Output: growth $e^{3N|\beta|}$ for every real angle and height.

Statement p. 20

Shift the periodic Fourier contour through arbitrary imaginary heights.

Interface

G.b. Only finitely many angular frequencies remain

Quarter-turn symmetry leaves $\ell \in 4\mathbb{Z}$ with $|\ell| \le 3N$.

Statement p. 21

Definitions/setup Model-specific Interface (general)

The result is local in the support configuration and global in the complex angle. It does not yet prove covariance. The positive forms in P.d and degree comparisons in E eliminate the surviving frequencies. The geometric reduction R is used only after that.

Full write-up estimate: about 2 pages.

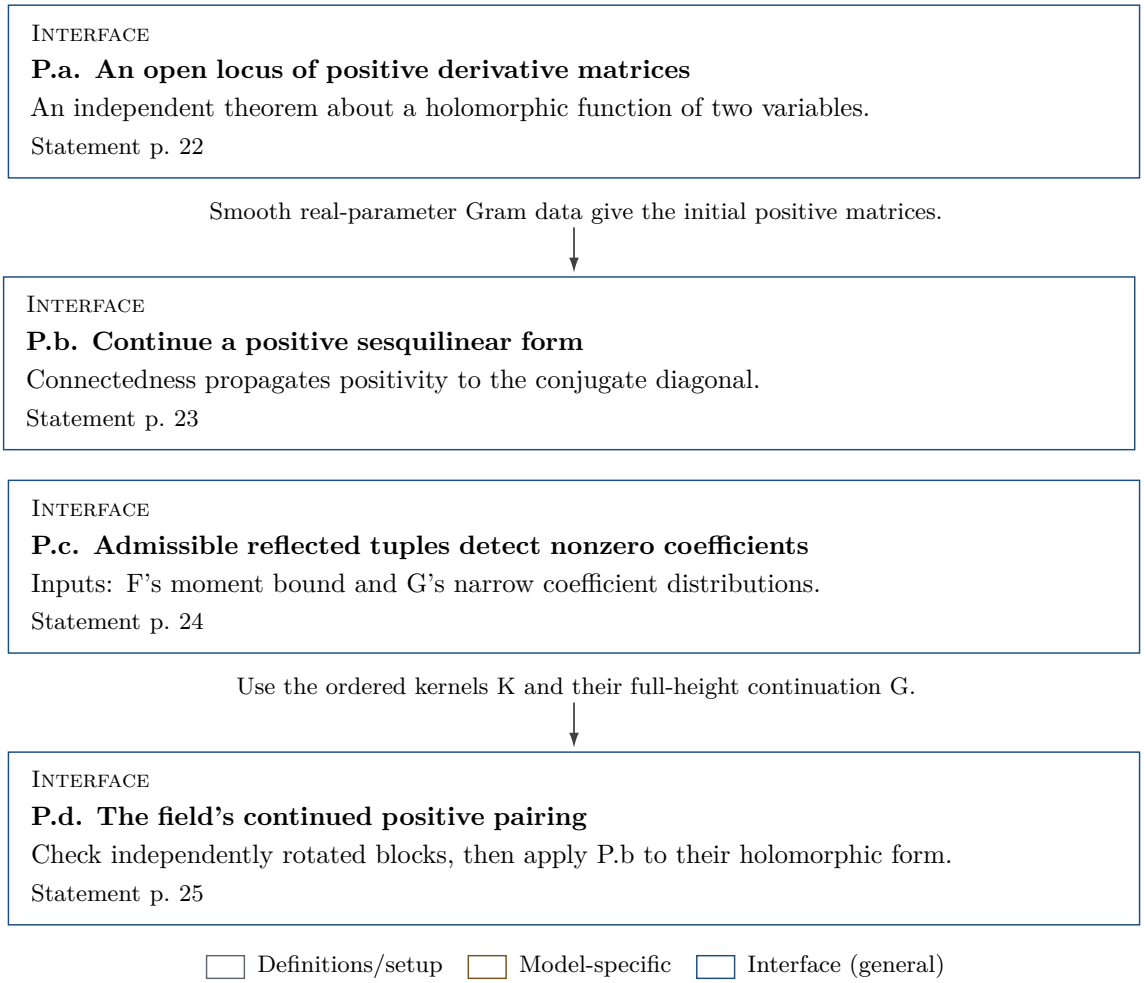
Source. RPM1, Proposition 7.1.

7. Holomorphic positivity and the continued reflected forms

P. Holomorphic positivity and the continued reflected forms

Interface. Overview Target.

The analytic theorem requires no field or exponential hypothesis. Its application uses the actual ordered kernels and explicit support conditions; an inequality alone is not analytically continued.



The first two tasks form the general analytic tool. The last two verify its field application and the tests needed by E. Empty tuples and finite families containing different degrees are included.

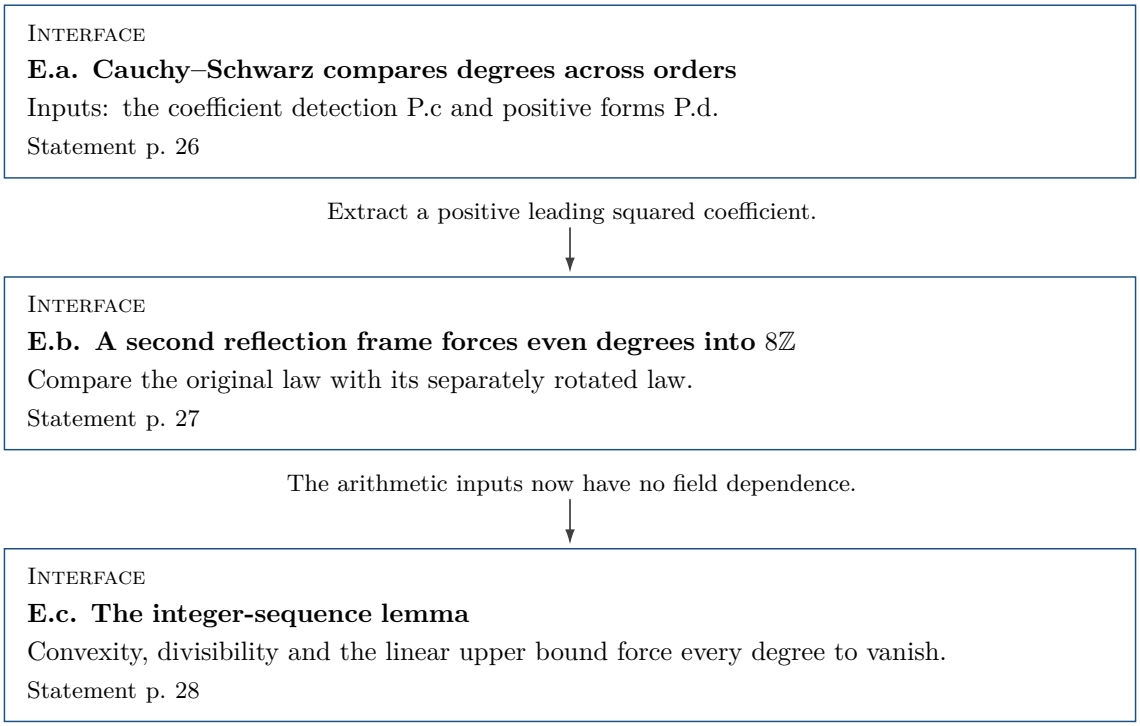
Full write-up estimate: about 4 pages.

Source. RPM1, Section 5, Section 7.2.

E. Positivity eliminates all remaining angular degrees

Interface. Overview Target.

Let M_N be the largest nonzero absolute Fourier degree occurring near *any* narrow N-point configuration, or zero if none occurs. Then G gives $M_0 = 0$ and $0 \le M_N \le 3N$, with $M_N \in 4\mathbb{Z}$.



The conclusion is $D_N = 0$ near all narrow configurations. This is the narrow theorem used by the geometric propagation R.b.

Full write-up estimate: about 2 pages.

Source. RPM1, Section 8.

9. From narrow configurations to all distinct points

R. From narrow configurations to all distinct points

Interface. Overview Target.

The geometric argument uses analytic translations of entire groups separated by coordinate cuts. This analytic input branches directly from the cone estimate C.b.

Interface
R.a. Analyticity under translations of one separated group
A spectral matrix element with a positive support margin.
Statement p. 29

Use the narrow covariance conclusion E.c.

Interface
R.b. The four-mirror planar covariance theorem
Narrowing and a compact-orbit argument remove all distinct-projection configurations.
Statement p. 30

Use additional coordinate frames for transverse separating cuts.

Interface
R.c. Remove projected coincidences outside the big diagonal
Induct on the number of coincident planar projections.
Statement p. 31

Definitions/setup Model-specific Interface (general)

For one selected plane, put

$$\Delta_2 = \{(z_1, \dots, z_N) : \pi z_j = \pi z_k \text{ for some } j \neq k\}, \quad \pi(t, x, y) = (t, x).$$

The four-mirror result stops outside Δ_2. For $d > 2$, the last step is needed to reach the complement of $\Delta = \{z_j = z_k \text{ for some } j \neq k\}$.

Full write-up estimate: 3–4 pages, excluding the cone and planar proof.

Source. RPM1, Lemma 2.1, Theorem B, Theorem A.

Appendix A. Detailed proof tasks and supporting statements

The following pages expand the principal results. Each states its own inputs and gives the source location of the argument.

F.a. Exponential field bounds and moment kernels

Interface. Overview Parent F Target.

Statement. Let μ be a Borel law on real tempered distributions valued in a finite-dimensional inner-product space. Assume $\mathbb{E}e^{F(\chi)} \le e^{C(\|\chi\|_1+\|\chi\|_2^2)}$ for every real Schwartz test and one $C > 0$. For $1 \le p < \infty$,

$$\|F(\chi)\|_{L^p(\mu)} \le K(C)(\|\chi\|_1 + \sqrt{p}\,\|\chi\|_2).$$

The field extends continuously to real $L^1 \cap L^2$ tests, with values in every finite L^p, using the norm $\|\chi\|_1 + \|\chi\|_2$. The extensions agree, and the real exponential bound remains valid. The moments obey

$$|S_N(\chi_1, \dots, \chi_N)| \le (K(C)\sqrt{N})^N \prod_j (\|\chi_j\|_1 + \|\chi_j\|_2), \qquad S_0 = 1.$$

They are continuous multilinear forms on $L^1 \cap L^2$ tests and tempered distributional kernels on Schwartz tests. The mean need not vanish. Translation invariance gives $S_1(\chi) = \langle m, \int \chi \rangle$ for a constant $m \in \mathcal{T}$. The four prescribed mirrors in a coordinate plane fix m under quarter turns; the weight restriction then makes m invariant under all rotations in that plane. Under A, this holds in every coordinate plane. Thus the one-point moment is covariant.

Proof to write. For real $\chi \ne 0$, optimize the exponential Markov bound for $\pm uF(\chi)$ to obtain

$$\mu\{|F(\chi)| > C\|\chi\|_1 + r\} \le 2\exp[-r^2/(4C\|\chi\|_2^2)], \qquad r > 0.$$

Integrate the tail, complete the field map in $L^1 \cap L^2$, and use Fatou for the extended exponential bound. Hölder gives the moment estimate; the Schwartz kernel theorem gives the distributional kernel. Translation invariance makes the first moment constant, and the quarter-turn fixed weights in $\{-2, -1, 0, 1, 2\}$ have weight zero.

Approximation needed later. The moment bound makes the characteristic-functional Taylor series absolutely convergent and makes $e^{iF(\chi)}$ limits of polynomials in $L^2(\mu)$. Their span is dense in the positive-side L^2 space: orthogonality gives zero Fourier transforms of each finite-dimensional conditional measure, then zero conditional expectations. This is the density input in C.a. The $L^1 \cap L^2$ extension permits compact planar averaging functions and transverse $L^1 \cap L^2$ profiles in I.b.

Source. RPM1, (144), (13), Appendix A. The first moment is treated in Section 1.1; polynomial density is in Section 4.1.

F.b. One RP mirror and translations imply reflection invariance

INTERFACE. Overview Parent F Target.

Independent statement. Let μ be a translation-invariant Borel law on real tempered fields with a fixed orthogonal tensor reflection action. Write $\Phi(\chi) = \mathbb{E}e^{iF(\chi)}$. Suppose that at a hyperplane H with reflection Θ, every finite real test family supported in one open side has positive semidefinite matrix $[\Phi(\chi_a - \Theta^*\chi_b)]_{a,b}$. Then

$$\Phi(\Theta^*\chi) = \Phi(\chi) \qquad (\chi \text{ every real Schwartz test}).$$

Thus the law is invariant under this reflection. An exponential bound is not required.

Proof to write. For a compact test on the chosen side, apply RP to $(\chi, 0)$:

$$\begin{pmatrix} \Phi(\chi - \Theta^*\chi) & \Phi(\chi) \\ \Phi(-\Theta^*\chi) & 1 \end{pmatrix} \geq 0.$$

Hermiticity and $\Phi(-\psi) = \overline{\Phi(\psi)}$ give the desired equality for this test. Translate an arbitrary compact test into the open side. The relation between reflection and normal translation, together with translation invariance, gives the equality for that test as well. Compact approximation in Schwartz space finishes the proof; continuity of Φ follows by dominated convergence.

Use in the rotation criterion. Apply the statement at every prescribed coordinate and diagonal plane. Coordinate reflections and coordinate exchanges generate the signed coordinate permutations. In particular quarter turns preserve the law. These symmetries justify the coordinate-frame changes in K.b and angular periodicity in G.b. The separate 45° change of law in E.b is not asserted to preserve this law.

Source. RPM1, Proposition B.1, Section 1.1.

C.a. The reflected Hilbert quotient and stationary translations

INTERFACE. Overview Parent C Target.

Statement and construction. Let μ be a translation-invariant law on real tempered fields with the exponential bound of F.a, and assume characteristic RP at $t = 0$. By F.b reflection preserves the law. Put $L^2_+ = L^2(\mu; \sigma\{F(\chi) : \operatorname{supp}\chi \subset \{t > 0\}\})$ and define

$$JA = \overline{A(\Theta^* F)}, \qquad \langle A, B\rangle_{\mathrm{OS}} = \mathbb{E}[(JA)B], \qquad \Theta(t, x, y) = (-t, x, y).$$

The form is positive semidefinite; quotienting its null space and completing gives $\mathcal{H}$, a contractive class map $A \mapsto [A]$, and vacuum $\Omega = [1]$. Compact positive-time field polynomials, including the constant, have dense classes.

For $\tau_a\chi(z) = \chi(z - a)$, put $U(a)F(\chi) = F(\tau_a\chi)$. There are strongly commuting self-adjoint generators $H \geq 0$ and P such that the induced translations are

$$V(s, x) = e^{-sH+ixP} \quad (s \geq 0), \qquad V(s, x)\Omega = V(s, x)^*\Omega = \Omega.$$

Proof to write. On finite sums of field exponentials, characteristic RP gives the positive form; the matrix transpose from the chosen convention remains positive. The bound $|\langle A, B\rangle_{\mathrm{OS}}| \leq \|A\|_{L^2}\|B\|_{L^2}$ and the density in F.a extend it to L^2_+.

Reflection and translation give

$$\|[U(s)A]\|^2 = \langle [A], [U(2s)A]\rangle \leq \|[A]\|\,\|[U(2s)A]\|.$$

Iterate and use the uniform ordinary L^2 bound to obtain contractivity and preservation of null vectors. Tangential translations are unitary on the quotient. Strong continuity holds first on field exponentials by the $L^1 \cap L^2$ test bound, then on the completion by density. The joint spectral theorem gives the displayed generators and their joint projection-valued measure. All norms of classes and insertions below refer to this quotient norm.

Source. RPM1, (39), (40), (42).

C.b. Comparing two spectral representations forces the cone

INTERFACE. Overview Parent C Target.

Independent measure criterion. Let ρ be a finite positive measure on $[0,\infty)\times\mathbb{R}$ and ρ_Δ a finite complex measure on the same space. Suppose, for all $u, v > 0$,

$$\int e^{(ip-\omega)v/2-(\omega+ip)u/2}\,d\rho(\omega,p) = \int e^{-Ev+iku}\,d\rho_\Delta(E,k).$$

Then ρ is supported on $\omega \geq |p|$.

Proof to write. Integrate against $h(v)\overline{h(u)}$ to get

$$\int \left|\mathcal{L}h((\omega-ip)/2)\right|^2 d\rho \leq \|\rho_\Delta\| \sup_{E\geq 0}|\mathcal{L}h(E)| \sup_{k\in\mathbb{R}}|\mathcal{L}h(ik)|.$$

Use $h(s) = s^{i\gamma}e^{-\varepsilon s}$. Up to a nonzero constant, its transform is $(1+z/\varepsilon)^{-1-i\gamma}$. After cancelling that constant, the right side grows at most as $e^{\pi|\gamma|/2}$. The positive integral on any sector $\arg(\varepsilon + (\omega-ip)/2) > \pi/4+\eta$ grows faster as $\gamma\to+\infty$, so that sector has zero mass. The negative probe excludes the opposite sector. Hence $|p| \leq \omega + 2\varepsilon$; send $\varepsilon \downarrow 0$.

Application to two RP frames. For the same stationary field as C.a, assume RP also at $t+x=0$. Move any real compact positive-time polynomial A into $t > |x|$ by a forward translation. Write J_0, J_Δ for the coordinate and diagonal antilinear reflections and $[\cdot]_0, [\cdot]_\Delta$ for their quotient maps. For $T = U(\delta t e_0 + \delta x e_1)$,

$$\mathbb{E}[(J_0A)TA] = \langle [A]_0, [TA]_0\rangle_0 = \langle [J_\Delta J_0 A]_\Delta, [TA]_\Delta\rangle_\Delta.$$

The first spectral representation has exponent $-\omega\delta t + ip\delta x$; the second has exponent $-E(\delta t+\delta x) + ik(\delta t - \delta x)$. Putting $v = \delta t+\delta x > 0$, $u = \delta t - \delta x > 0$ gives exactly the measure identity above. The first measure is positive; the second has finite variation by the quotient Cauchy–Schwarz bound. The preliminary translation multiplies the first spectral measure by $e^{-2b\omega} > 0$, preserving its support. Density gives $H \geq |P|$ on all of $\mathcal{H}$.

The joint spectral calculus now defines $V(z) = e^{-z_tH+iz_xP}$, contractive for $q(z) = \Re z_t - |\Im z_x| \geq 0$, norm holomorphic for $q > 0$. On $q \geq \delta > 0$, a derivative of total order k is bounded by $k!\delta^{-k}$. This is the analytic input in I.b and R.a.

Source. RPM1, Proposition 4.1, (75), (76).

I.a. A slab source and a bounded field insertion

INTERFACE. Overview Parent I Target.

Statement. Use a translation-invariant, t-reflection-positive field with the bound of F.a, its positive quotient $\mathcal{H}$, classes $[A]$ and translations from C.a. Here extend F complex-linearly to $(L^1 \cap L^2)(\mathbb{R}^d; \mathcal{T})_{\mathbb{C}}$. For a source h in this space supported in $0 \le t \le l$ and a bounded positive-time observable A,

$$\|[e^{F(h)} U(le_0)A]\| \le e^{C(\|\Re h\|_1 + \|\Re h\|_2^2)} \|[A]\|.$$

For χ supported in $|t| \le \delta$, this gives a well-defined bounded operator

$$B_\delta(\chi)[A] = [F(\tau_{\delta e_0}\chi) U(2\delta e_0)A], \qquad \|B_\delta(\chi)\| \le K(\|\chi\|_1 + \|\chi\|_2).$$

Here τ_a moves test supports by a and the norm is the OS quotient norm. Field products are allowed by L^4 approximation.

Proof to write. Let $Jh = \Theta^* \bar{h}$ and duplicate the slab source by $h_{r+1} = \tau_{2^r l e_0}(Jh_r + h_r)$, $h_0 = h$. Adjacent slab interiors are disjoint, so $\|\Re h_r\|_1 = 2^r \|\Re h\|_1$ and $\|\Re h_r\|_2^2 = 2^r \|\Re h\|_2^2$. For $a_r = \|[e^{F(h_r)} U(2^r l e_0)A]\|$, reflection gives

$$a_r^2 \le \|[A]\| a_{r+1}, \qquad a_0 \le \|[A]\|^{1-2^{-r}} a_r^{2^{-r}}.$$

The exponential bound gives

$$a_r^2 \le \|A\|_\infty^2 e^{C 2^{r+1}(\|\Re h\|_1 + \|\Re h\|_2^2)}.$$

Let $r \to \infty$. Differentiate the entire Hilbert-valued source map on $|z| = (\|\chi\|_1 + \|\chi\|_2)^{-1}$ for $\chi \ne 0$ using Cauchy's estimate. The bound annihilates null vectors, making B_δ a quotient operator.

Identities exported to planar localization. With $J\chi = \Theta^* \bar{\chi}$, reflection gives $B_\delta(\chi)^* = B_\delta(J\chi)$. For $r \ge |t|$,

$$V(r+t, x) B_\delta(\chi) V(r-t, -x) = B_{\delta + r}(\tau_{(t,x,0)}\chi).$$

If χ is supported in $|t - c| \le \delta_0$ and $|c| + \delta_0 \le \delta$, the same identity gives

$$B_\delta(\chi) = e^{-(\delta + c - \delta_0)H} B_{\delta_0}(\tau_{-ce_0}\chi) e^{-(\delta - c - \delta_0)H}.$$

Source. RPM1, (80), (86), Section 6.2.

I.b. Planar localization without removing transverse smearing

INTERFACE. Overview Parent I Target.

Statement. Use the operators B_δ from I.a and the contractive holomorphic $V(z)$ on $\mathcal{W} = \{q(z) > 0\}$ from C.b. Fix $f \in (L^1 \cap L^2)(\mathbb{R}^{d-2}; \mathcal{T})_{\mathbb{C}}$, including $d = 2$ with no transverse variable; then both profile norms are $|f|$. Define

$$j_\varepsilon(z) = (\pi\varepsilon^2)^{-1} 1_{|z|<\varepsilon}, \qquad A_{f,\varepsilon}(\xi,\eta) = V(\xi - \varepsilon e_0) B_\varepsilon(j_\varepsilon f) V(\eta - \varepsilon e_0).$$

These regularized operators are defined for $0 < \varepsilon < q_0 = \min(q(\xi), q(\eta))$. For $q_0 > 0$ they have a unique weak operator limit $A_f(\xi,\eta)$ as $\varepsilon \downarrow 0$. It is linear in f, norm holomorphic in (ξ,η) and satisfies

$$\|A_f(\xi,\eta)\| \le K \left(\|f\|_1 + \frac{\|f\|_2}{\min(q(\xi), q(\eta))} \right).$$

The fixed-gap estimate and unique limit. Put $q_0 = \min(q(\xi), q(\eta))$ and $r = (q_0 - \varepsilon)/2$. For matrix elements of $A_{f,\varepsilon}(\xi + \zeta, \eta - \zeta)$, translation covariance and I.a bound integration against $\alpha \in L^2((-r,r)^2)$ by $K(2r\|f\|_1 + \|f\|_2)\|\alpha\|_2$, since $\|\alpha\|_1 \le 2r\|\alpha\|_2$. Duality and the holomorphic mean-value inequality give $\|A_{f,\varepsilon}\| \le K(\|f\|_1 + (q_0 - \varepsilon)^{-1}\|f\|_2)$.

Local scalar normal-family compactness follows. Fix $\delta = q_0/2$, take $0 < \varepsilon < \delta/2$, and let $\alpha \in L^2(\mathbb{R}^2)$ be supported in $(-\delta/2, \delta/2)^2$. The identity

$$\int \alpha(z) A_{f,\varepsilon}(\xi + z, \eta - z)\, dz = V(\xi - \delta e_0) B_\delta((\alpha * j_\varepsilon) f) V(\eta - \delta e_0)$$

converges in operator norm because $\alpha * j_\varepsilon \to \alpha$ in both L^1 and L^2. It identifies all scalar subsequential limits on a real square; holomorphic uniqueness identifies them everywhere. The resulting bounded forms define the unique weak limit. Shift factorization then proves norm holomorphy.

Identities required by the kernel construction.

$$V(\kappa) A_f(\xi,\eta) V(\kappa') = A_f(\kappa + \xi, \eta + \kappa'), \qquad q(\kappa), q(\kappa') \ge 0.$$

Writing $Jf = \rho(\Theta)\bar{f}$, the adjoint is $A_f(\xi,\eta)^* = A_{Jf}(-\Theta\bar{\eta}, -\Theta\bar{\xi})$. For compactly supported $\alpha \in L^2(\mathbb{R}^2)$ with $|t| < \delta$,

$$B_\delta(\alpha f) = \int \alpha(z) A_f(\delta e_0 + z, \delta e_0 - z)\, dz.$$

Finally, $A_f(\xi, a) A_g(b, \eta)$ depends on $a, b \in \mathcal{W}$ only through $a + b$: subtract a small common δe_0 and apply the shift identity. These are exact operator identities.

Source. RPM1, (91), (95), (96), Section 6.3.

K.a. An ordered kernel representing the original moments

INTERFACE. Overview Parent K Target.

Statement and formula. Use $\mathcal{H}, V, \Omega$ and A_f with the norm, shift and smearing identities of I.b. Let $N \geq 2$, $Z = (z_1, \dots, z_N)$, $\Delta_j = z_{j+1} - z_j \in \mathcal{W}$, and $f_j \in (L^1 \cap L^2)(\mathbb{R}^{d-2}; \mathcal{T})_{\mathbb{C}}$. Define

$$S_N(Z; f) = \langle \Omega, A_{f_1}(\kappa, \Delta_1/2) \Big[\prod_{j=2}^{N-1} A_{f_j}(\Delta_{j-1}/2, \Delta_j/2) \Big] A_{f_N}(\Delta_{N-1}/2, \kappa)\Omega \rangle.$$

The middle product is ordered by increasing j and empty for $N = 2$. Any $\kappa \in \mathcal{W}$ gives the same value. The kernel is holomorphic, translation invariant and multilinear in f. It obeys

$$|S_N(Z; f)| \leq K^N \prod_j (\|f_j\|_1 + q_*^{-1} \|f_j\|_2), \qquad q_* := \min_{j<N} q(\Delta_j).$$

For compact planar α_j whose supports lie in strictly ordered time slabs,

$$S_N(\alpha_1 f_1, \dots, \alpha_N f_N) = \int \prod_j \alpha_j(z_j) S_N(Z; f)\, dZ.$$

Set $S_0 = 1$ and $S_1(z; f) = \langle m, \int f \rangle$, with the constant m from F.a; for $d = 2$, this is $\langle m, f \rangle$.

Proof to write. The shift identity and $V\Omega = V^*\Omega = \Omega$ remove the choice of κ. Take it with a sufficiently large time component and apply the insertion norm estimate to obtain the gap bound.

For the smearing identity, enclose the compact supports in disjoint time slabs, then translate them to positive time. The definition of B_δ and its slab-localization formula express the genuine product-of-fields class as a product of B_δ operators and intervening positive-time semigroups. Substitute the exact smearing identity of I.b. Adjacent semigroups combine; the inner-gap identity assigns one half-gap to each neighboring insertion. Pair with the vacuum. The gaps are positive uniformly on the compact supports, so the integral is bounded and equals the original moment. This identification is needed both for gluing and for real-angle identification of the angular continuation.

Source. RPM1, (104), (105), (106).

K.b. Gluing permuted and quarter-turned ordering domains

INTERFACE. Overview Parent K Target.

Statement. Suppose K.a's ordered kernels represent one symmetric moment functional, and suppose quarter turns in the selected plane preserve the law and act orthogonally on the profiles. Such symmetry follows from F.b under the four planar RP hypotheses. Permute the ordering and use each quarter-turned coordinate frame. The resulting holomorphic kernels agree on overlaps and give one kernel on their union. It is symmetric under simultaneous point and profile permutations and satisfies

$$S_N(R(-\pi/2)Z; \varrho(\pi/2)f) = S_N(Z; f), \qquad \varrho(\zeta) = \rho(R(\zeta)).$$

Its real domain includes configurations with all times distinct or all x coordinates distinct. The same mixed bound of K.a holds in each chosen ordering frame.

Proof to write. On a quarter-turned domain, rotate the points to the original ordered domain and apply inverse tensor pullback to the profiles. Each domain is defined by strict linear inequalities on real and imaginary parts. Any nonempty intersection is convex and contains the real part of each of its points. It therefore contains a real open subset.

On that real subset, both kernels give the same original moments after integration against products of compact scalar tests. The smearing identity in K.a identifies them there. Holomorphic uniqueness identifies them throughout the connected intersection, so the kernels glue. Permutation symmetry and the quarter-turn identity follow from their construction and the same uniqueness.

Use. The full-height angular charts in G.a use precisely these ordering frames. The independently rotated blocks in P.d also use this joined kernel. No assertion of joint analyticity at every distinct full spacetime configuration is needed.

Source. RPM1, Section 6.5, (108).

G.a. A full-height angular continuation and its growth bound

INTERFACE. Overview Parent G Target.

Statement. Use the joined kernels of K, and assume their planar tensor weights satisfy $|m| \le 2$. Fix narrow compact planar supports, an ordering and a common direction $n = R(\phi)e_0$ with half-angle $\alpha < \pi/4$. For product tests $\chi_j(t,x,y) = \alpha_j(t,x)f_j(y)$ in these supports, with $\alpha_j \in C_c^\infty(\mathbb{R}^2;\mathbb{R})$ and transverse $L^1 \cap L^2$ profiles, the actual rotation defect has an entire, 2π-periodic continuation with

$$|D_N[\chi](\theta + i\beta)| \le C_\chi e^{3N|\beta|} \qquad (\theta, \beta \in \mathbb{R}).$$

Construct and identify every angular chart. For each integer k, use the vertical strip $|\Re\zeta - \phi - k\pi/2| < \pi/2 - \alpha$ and define

$$\int \prod_j \alpha_j(z_j) S_N(R(-\zeta + k\pi/2)Z; \varrho(\zeta - k\pi/2)f)\, dZ - S_N(\chi).$$

The scalar tests remain at real arguments. In the nearest quarter-turn frame every real gap has a common lower bound $\delta > 0$ in q, uniformly over the support and real angle. The identities for $z_t \pm iz_x$ give

$$q(R(-i\beta)z) \ge e^{-|\beta|} q(z) \quad (q(z) > 0).$$

Thus each strip extends through every imaginary height. Its real values are the actual defect by quarter-turn symmetry and the exact smearing identity. The strips overlap in real intervals and cover $\mathbb{C}$; uniqueness glues them. Shifting k by four proves periodicity.

The quantitative estimate. The rotated tuple satisfies $q_*^{-1} \le \delta^{-1} e^{|\beta|}$. The weight restriction gives $\|\varrho(\theta + i\beta - k\pi/2)f_j\|_p \le e^{2|\beta|}\|f_j\|_p$ for both $p = 1, 2$. Each factor in K's bound is therefore at most $e^{3|\beta|}(\|f_j\|_1 + \delta^{-1}\|f_j\|_2)$. One may take

$$C_\chi = K^N \prod_j [\|\alpha_j\|_1(\|f_j\|_1 + \delta^{-1}\|f_j\|_2)] + |S_N(\chi)|.$$

It is independent of both θ and β; the angular exponent remains $3N$. The defects D_0 and D_1 are already zero by F.a.

Source. RPM1, Proposition 7.1, (114), (115).

G.b. A periodic growth bound gives finite Fourier degree

INTERFACE. Overview Parent G Target.

Independent analytic statement. Let h be entire and 2π-periodic, with $|h(\theta+i\beta)| \le Ae^{q|\beta|}$ for all real θ, β, where A, q are finite and nonnegative. Then

$$h(\zeta) = \sum_{|\ell|\le q} c_\ell e^{i\ell\zeta}.$$

If h has period $\pi/2$, only $\ell \in 4\mathbb{Z}$ occur. If also $h(0) = 0$, this becomes

$$h(\zeta) = \sum_{\substack{\ell\in 4\mathbb{Z} \\ 0<|\ell|\le q}} c_\ell (e^{i\ell\zeta} - 1).$$

Proof to write. The periodic rectangular contour gives, at every real height β,

$$c_\ell = \frac{1}{2\pi}\int_0^{2\pi} h(\theta+i\beta)e^{-i\ell\theta}e^{\ell\beta}\,d\theta, \qquad |c_\ell| \le Ae^{q|\beta|+\ell\beta}.$$

Choose $\beta \to -\infty$ for $\ell > q$ and $\beta \to +\infty$ for $\ell < -q$. Fourier uniqueness and holomorphic uniqueness give the finite expansion. The shorter period restricts the remaining integers, and $h(0) = 0$ fixes the constant term.

Application to moments. Apply the statement with $h = D_N[\chi]$ and $q = 3N$, using G.a. Quarter turns give the shorter period by F.b. These are the original real-angle Fourier coefficients,

$$c_\ell^{(N)}[\chi] = \frac{1}{2\pi}\int_0^{2\pi} D_N[\chi](\theta)e^{-i\ell\theta}\,d\theta.$$

By F.a they are bounded multilinear forms on $L^1 \cap L^2$ tests. Product density therefore extends the finite-defect formula to all tests with the same small compact planar supports. It applies on the narrow configurations used in P.c and E.

Source. RPM1, (116), (117).

P.a. The positive derivative-matrix locus is open

INTERFACE. Overview Parent P Target.

Independent theorem. Let $O \subset \mathbb{C}$ be open and invariant under complex conjugation, and let $k : O^2 \to \mathbb{C}$ be holomorphic. For each $N \geq 0$ put

$$M_N(z)_{a,b} = \partial_\zeta^a \partial_\xi^b k(\zeta, \xi)\Big|_{(z,\bar{z})}, \qquad 0 \leq a, b \leq N.$$

Then

$$\Omega_k = \{z \in O : M_N(z) \text{ is positive semidefinite for every } N\}$$

is open in O. No connectedness, distinguished origin, field, reflection or Gram representation is assumed.

Proof to write. Fix $z_0 \in \Omega_k$ and a polydisc centered at $(z_0, \bar{z}_0)$ compactly contained in O^2. Taylor expansion gives, for z in a smaller common disc,

$$M_N(z) = \lim_{L \to \infty} Z_L M_{N+L}(z_0) Z_L^*,$$

where Z_L is an $(N+1)$ by $(N+L+1)$ matrix with

$$(Z_L)_{a,c} = \begin{cases} (z - z_0)^{c-a}/(c-a)!, & 0 \leq c - a \leq L, \\ 0, & \text{otherwise.} \end{cases}$$

Each finite matrix is a positive congruence. Its entries converge absolutely to the mixed derivatives defining $M_N(z)$, so the limit is positive semidefinite.

The same polydisc works for every derivative order N: all derivatives are holomorphic on that polydisc. This gives one neighborhood of z_0 on which *all* derivative matrices are positive, as required.

Use. Continuity makes Ω_k closed. In P.b, smooth real-parameter Gram data put the origin in Ω_k; this theorem supplies the openness step in the connected-domain argument.

Source. RPM1, Theorem D, (71).

P.b. Smooth Gram data determine a positive holomorphic continuation

INTERFACE. Overview Parent P Target.

Independent theorem. Let X be a complex vector space, $\mathcal{H}$ a complex Hilbert space, and $O \subset \mathbb{C}$ a connected conjugation-invariant open set containing zero. Suppose $K_{\zeta,\xi}$ are sesquilinear forms on X, antilinear in the first entry, such that $K_{\zeta,\xi}(A,B)$ is jointly holomorphic on O^2 for every A, B.

For every $A \in X$, assume there is a smooth curve $v_A : [0, \delta_A) \to \mathcal{H}$ and that, for all small real $r, s \geq 0$ and every pair A, B,

$$K_{r,s}(A,B) = \langle v_A(r), v_B(s) \rangle_{\mathcal{H}}.$$

Then $K_{z,\bar{z}}$ is positive semidefinite on X for every $z \in O$. Only finite families in X enter this assertion.

Proof to write. For fixed A, let $k(\zeta,\xi) = K_{\zeta,\xi}(A,A)$. The mixed derivative matrices at zero are Gram matrices:

$$(M_N(0))_{a,b} = \langle v_A^{(a)}(0), v_A^{(b)}(0) \rangle.$$

One-sided derivatives suffice. Hence $0 \in \Omega_k$. By P.a and continuity, Ω_k is open and closed in connected O; therefore it is all of O. The zeroth entry gives $K_{z,\bar{z}}(A,A) \geq 0$. Apply this to every finite linear combination of the entries of a chosen family. Sesquilinearity gives the stated positive semidefinite form.

Field application. The field construction supplies smooth curves in L^2_+ and a continuous positive semidefinite reflection form. Applying the contractive quotient map of C.a gives the Hilbert-space curves required here. The substantive application check is P.d.

Source. RPM1, Lemma 5.1, (64).

P.c. Admissible reflected tuples and detection of coefficients

INTERFACE. Overview Parent P Target.

Support conditions and statement. Let $\mathcal{C}_+ = \{(t,x) : t > |x|\}$. A real product tuple $\chi_j = \alpha_j(t,x)f_j(y)$, with $\alpha_j \in C_c^\infty(\mathbb{R}^2;\mathbb{R})$, is admissible when its compact planar supports K_j satisfy

$$K_j \subset \mathcal{C}_+, \qquad K_{j+1} - K_j \subset \mathcal{C}_+.$$

Use real transverse $L^1 \cap L^2$ profiles and include the empty tuple. For any two admissible tuples χ, η, their reflected concatenation $(J\chi, \eta)$ is narrow. This holds for mixed pairings and for both squared tuples, even if the positive tuples overlap each other.

For the coefficients $c_\ell^{(N)}$ in G.b, if $\ell \neq 0$ is nonzero near a narrow configuration, then for *every* split $a + b = N$ there are admissible tuples of lengths a, b with

$$c_\ell^{(N)}[J\chi, \eta] \neq 0.$$

Here J reflects and conjugates the first tuple; reverse its order when listing the concatenated planar points.

Proof to write. The reflected block's successive gaps are the original ones acted on by $-\Theta$, which preserves $\mathcal{C}_+$. The crossing gap is $z_1' - \Theta z_1$, a sum of two elements of $\mathcal{C}_+$. Thus all gaps, including the crossing, remain in this cone.

The defect cocycle gives $c_\ell[R(\phi)^*\chi] = e^{i\ell\phi}c_\ell[\chi]$ for $\ell \neq 0$. Common translations and label permutations also preserve nonvanishing. Rotate a narrow direction to time and put the origin at the midpoint of the chosen $a|b$ gap. Each vector to this midpoint is a half-gap plus a sum of cone gaps. Reflect and reverse the first block; both blocks are admissible in small product neighborhoods. If one block is empty, move the origin sufficiently far along time instead.

The distributional coefficient is nonzero on a product test in these neighborhoods. Product density in $L^1 \cap L^2$ and the moment bound in F.a give the displayed real product tests. This verifies the support class used by every Cauchy–Schwarz comparison in E.

Source. RPM1, Lemma 7.2, (119), (120).

P.d. The actual continued reflection pairing

INTERFACE. Overview Parent P Target.

Statement and construction. Use the field kernels K, the narrow continuation G, and the admissible product tuples in P.c. Take formal finite linear combinations of monomials $A = \prod_{j=1}^{a} F(\chi_j)$, including the vacuum. For a fixed finite family there is a vertical strip $O = \{z : |\Re z| < \delta\}$ and jointly holomorphic sesquilinear forms $K_{\zeta,\xi}$ on O^2 with

$$K_{r,s}(A,B) = \langle A(r), B(s)\rangle_{\mathrm{OS}}, \qquad A(r) = \prod_j F(R(r)^*\chi_j)$$

for small real r, s. At every real β,

$$\langle A, B\rangle_\beta := K_{-i\beta,i\beta}(A,B) = S_{a+b}(J\chi,\eta) + D_{a+b}[J\chi,\eta](i\beta)$$

is a positive semidefinite form. The first tuple has length a and the second length b. Formal monomials avoid requiring an arbitrary representation of an L^2 observable to define the continuation.

Verify the holomorphic form. Integrate the ordered kernel against the fixed real scalar tests. Rotate the first reflected block and its profiles as

$$R(\zeta)\Theta Z, \quad \varrho(-\zeta)Jf, \qquad \text{and the second as} \qquad R(-\xi)Z', \quad \varrho(\xi)g.$$

Choose δ so small that all real rotations in the strip keep each positive support and every successive within-block displacement strictly in $\mathcal{C}_+$. Independent imaginary rotations preserve $\mathcal{W}$. The reflected block lies in $-\mathcal{W}$, the second in $\mathcal{W}$, and their crossing displacement is a sum of two elements of $\mathcal{W}$. The kernel bounds are uniform on compact subsets of O^2, proving joint holomorphy after integration.

For small real parameters the exact smearing identity gives the displayed OS Gram form. The field bounds make these curves smooth in L^2_+. Apply P.b. At $(-i\beta, i\beta)$ the two blocks have undergone one common imaginary rotation. Uniqueness and the finite narrow continuation identify the form with the displayed continued moment. This proves positivity for the same continuation whose Fourier coefficients are used in E.

Source. RPM1, Section 7.2, (124), (127).

E.a. A cross-order inequality for the maximal angular degree

INTERFACE. Overview Parent E Target.

Statement. Use G's finite narrow Fourier expansion, P.c's detection for every split, and P.d's positive continued forms. Define M_N as the maximum absolute nonzero coefficient degree over all local narrow test tuples of order N, setting it to zero if there is none. Then for all integers $a, b \geq 0$,

$$2M_{a+b} \leq M_{2a} + M_{2b}.$$

The same maximum controls all relevant mixed and squared tests; it is not a maximum for a single fixed tuple.

Proof to write. If $M_{a+b} > 0$, conjugation exchanges its positive and negative degrees. The testing statement supplies admissible tuples of lengths a, b such that $c_{-M_{a+b}}[J\chi, \eta] \neq 0$. For their monomials A, B, the finite expansion gives

$$\langle A, B\rangle_\beta = (c_{-M_{a+b}}[J\chi, \eta] + o(1))e^{M_{a+b}\beta} \quad (\beta \to +\infty),$$

while the squared terms are $O(e^{M_{2a}\beta})$ and $O(e^{M_{2b}\beta})$. By positivity,

$$|\langle A, B\rangle_\beta|^2 \leq \langle A, A\rangle_\beta \langle B, B\rangle_\beta.$$

The nonzero leading coefficient forces the exponent inequality. If $M_{a+b} = 0$ the result is immediate. Empty tuples include $a = 0$ or $b = 0$, a case needed by E.c.

All three concatenations are narrow by P.c; therefore the same finite-degree control applies to the mixed term and both squared terms. No information about coefficients outside the narrow locus is required.

Source. RPM1, (130), (131), (132).

E.b. The 45-degree comparison and divisibility by eight

INTERFACE. Overview Parent E Target.

Statement. For a field satisfying the four selected planar RP assumptions, the maximal narrow degrees of E.a obey

$$M_{2a} \in 8\mathbb{Z} \qquad (a \geq 0).$$

The comparison uses a separately rotated law and its own OS space; it does not assume invariance of the original law under 45°.

First obtain a strictly positive diagonal coefficient. Let $M = M_{2a} > 0$. On any finite family of admissible degree-a monomials, multiply the positive form of P.d by $e^{-M\beta}$ and let $\beta \to +\infty$. Its limit is the positive semidefinite form

$$(A, B) \longmapsto c_{-M}[J\chi, \eta].$$

Detection gives a nonzero mixed value. Cauchy–Schwarz for this limiting form then gives a tuple with $c_{-M}[J\chi, \chi] > 0$. The sign belongs to this leading squared-norm coefficient; arbitrary mixed coefficients have no such sign.

Compare the two laws. Set $Q = R(\pi/4)$ and define $\widetilde{\Phi}(\chi) = \Phi(Q^*\chi)$. This change permutes the four planar mirrors and preserves the exponential and translation hypotheses. Reconstruct the positive continued forms for $\widetilde{\mu}$ using exactly these four mirrors. The defect cocycle gives, for $\ell \in 4\mathbb{Z} \setminus \{0\}$,

$$\widetilde{c}_\ell[\chi] = e^{i\ell\pi/4} c_\ell[\chi] = (-1)^{\ell/4} c_\ell[\chi], \qquad \widetilde{M}_N = M_N.$$

The same admissible squared tuple must therefore satisfy

$$0 \leq \widetilde{c}_{-M}[J\chi, \chi] = (-1)^{M/4} c_{-M}[J\chi, \chi], \qquad c_{-M}[J\chi, \chi] > 0.$$

Thus $M/4$ is even. The case $M = 0$ is automatic. Other mirror families in dimension $d > 2$ are not needed by this comparison.

Source. RPM1, (134), (137), (139).

E.c. An independent integer-sequence lemma

INTERFACE. Overview Parent E Target.

Statement. Let $(M_n)_{n\ge 0}$ be nonnegative integers with $M_0 = 0$ and

$$M_n \le 3n, \qquad 2M_{a+b} \le M_{2a} + M_{2b}, \qquad M_{2n} \in 8\mathbb{Z}$$

for all nonnegative integers a, b, n. Then $M_n = 0$ for every n. No field, analytic continuation or RP hypothesis occurs here.

Proof to write. First $M_2 \le 6$ and $M_2 \in 8\mathbb{Z}$ imply $M_2 = 0$. Put $\delta_n = M_{2n} - M_{2n-2}$. Choosing $a = n-1$, $b = n+1$ in the cross-order inequality gives

$$2M_{2n} \le M_{2n-2} + M_{2n+2}, \qquad \delta_n \le \delta_{n+1}.$$

These increments are multiples of eight and start at $\delta_1 = 0$. They are nonnegative. If any $\delta_r > 0$, every later increment is at least eight, so

$$M_{2j} \ge 8(j - r + 1) \qquad (j \ge r).$$

This contradicts $M_{2j} \le 6j$ for large j. All even terms vanish. The inequality with $a = 0, b = n$ then gives $0 \le 2M_n \le M_0 + M_{2n} = 0$.

Close the narrow covariance argument. G supplies the bound $3N$; E.a supplies the cross-order inequality; E.b supplies the even-order divisibility. Hence every nonzero Fourier coefficient vanishes on narrow configurations. The defect is then constant in the angle and vanishes at angle zero, so it vanishes there for every real angle. This is the sole covariance input used by R.b.

Source. RPM1, Lemma 8.1. Its concluding application is in Section 9.

R.a. Analyticity under translations of one separated group

INTERFACE. Overview Parent R Target.

Statement. Use a stationary field satisfying F.a and a coordinate OS frame whose joint generators satisfy $H \geq |P_j|$ for each relevant tangential coordinate separately. These inequalities are supplied by C.b in the corresponding coordinate-diagonal planes. Let two products A_-, A_+ have compact supports in $t < -\delta$ and $t > \delta$, with $\delta > 0$. Translate only the positive group by $se_0 + v$, with v tangential. The smeared moment is a spectral matrix element

$$\mathbb{E}[A_- U(se_0 + v) A_+] = \langle \psi_-, V(2\delta + s, v)\psi_+ \rangle, \quad \psi_- = [U(-\delta e_0) J A_-], \quad \psi_+ = [U(-\delta e_0) A_+].$$

It is holomorphic for

$$2\delta + \Re s > \sum_j |\Im v_j|.$$

Thus it is real analytic for $s > -2\delta$ and every real tangential translation. Under only the four planar RP assumptions, this assertion uses t or x as time and the other planar coordinate as tangent. Under the full mirror hypotheses it holds for every coordinate normal.

Proof and applicability to local configurations. Reflection and stationarity give the matrix element. Products lie in L^2 by F.a. The separate spectral bounds $|p_j| \leq \omega$ make the joint exponential contractive in the displayed ℓ^1 tube; spectral derivative estimates give holomorphy there. This uses no Euclidean bound on the joint momentum norm.

Near a finite distinct-point configuration, choose a coordinate hyperplane containing none of its points. Localize the supports so each group stays a fixed distance from the plane, then apply the result. In ordinary pullback notation the moving-group tests are translated by $T^*_{-se_0 - v}$, which moves their supports forward. Empty groups give constant moments by stationarity.

Use. The proof needs analyticity of these whole-group moves to propagate zeros. It does not assert joint real analyticity in every point variable. These are the moves used by R.b and R.c.

Source. RPM1, Lemma 2.1, Section 6.1, (78).

R.b. Four planar RP mirrors give covariance at distinct projections

INTERFACE. Overview Parent R Target.

Standalone planar theorem. Let $d \geq 2$ and let μ be a stationary Borel law on real $\mathcal{T}$-valued tempered fields with $\mathbb{E}e^{F(\chi)} \leq e^{C(\|\chi\|_1+\|\chi\|_2^2)}$ for all real Schwartz tests. Suppose the orthogonal coefficient action has planar weights $|m| \leq 2$ in the selected (t, x) plane, and assume characteristic RP at

$$t = 0, \quad x = 0, \quad t + x = 0, \quad t - x = 0.$$

Then, for every planar angle θ, the distribution $D_N(\theta) = S_N \circ R(\theta)^* - S_N$ vanishes outside

$$\Delta_2 = \{\pi z_j = \pi z_k \text{ for some } j \neq k\}, \qquad \pi(t, x, y) = (t, x).$$

Equivalently it vanishes on tests localized near every configuration with distinct planar projections. The four mirrors alone suffice.

Proof from the two established inputs. E.c gives covariance near every narrow configuration; R.a gives the planar separated-group analyticity. First remove equal times: if two planar projections differ but their times agree, an x-cut separates them. Move one group a small amount in time. This reduces the number of equal-time pairs, preserving the x separation and creating no new equality. Both terms of the rotation defect are analytic along this move for all sufficiently small angles.

Once times are distinct, cut between successive times and move each later group forward until every successive time gap exceeds the absolute x difference. The result is narrow. Each of the finitely many paths keeps a positive separating gap; compactness makes it persist under all sufficiently small rotations and on small product neighborhoods. Analytic uniqueness propagates narrow vanishing back along these paths.

For every configuration on the compact planar rotation orbit this gives local small-angle covariance. A finite orbit cover gives one positive angle bound after shrinking the initial neighborhood. The cocycle

$$D_N[\chi](\theta_0 + \theta) = D_N[R(\theta_0)^*\chi](\theta) + D_N[\chi](\theta_0)$$

then propagates the equality to every angle. This completes the four-mirror theorem without claiming covariance on $\Delta_2 \setminus \Delta$.

Source. RPM1, Theorem B, Proposition 2.2, Lemma 2.3, Theorem C.

R.c. Transverse cuts remove coincidences of planar projections

INTERFACE. Overview Parent R Target.

Distributional implication. Let S_N be a tensor-valued distribution on $(\mathbb{R}^d)^N$, and fix a rotation R of the first two coordinates. Suppose its defect $D_R = S_N \circ R^* - S_N$ vanishes outside Δ_2. Assume the separated-group analyticity of R.a for every transverse coordinate cut: near any distinct-point configuration, translating all points on one side of such a cut by a small (t, x) vector gives real-analytic smeared values of S_N. Then D_R vanishes outside the full big diagonal Δ. For $d = 2$ this conclusion already follows from $\Delta_2 = \Delta$.

Proof to write. Induct on the number of pairs whose planar projections agree. If two such points are distinct, they differ in a transverse coordinate. Choose a transverse coordinate cut separating them and avoiding every point of the finite configuration. Move just one side by a sufficiently small nonzero time displacement. The selected projected coincidence disappears and no new one is created. Full points remain distinct.

The original and rotated moments are both analytic along this move. Indeed, the cut is transverse, so the planar rotation preserves it and changes only the planar displacement vector of its moving group. The induction hypothesis gives local vanishing at the moved configurations; analytic uniqueness propagates it to the original one. The finite induction removes all projected coincidences away from Δ.

Application and final assembly. Under A's full mirror assumptions, R.b gives the vanishing outside Δ_2 in each chosen plane, and C.b in the other coordinate frames gives the full cut-analyticity input R.a. Repeat this implication in every coordinate plane. These planar rotations generate $SO(d)$; F.b's coordinate reflections give $O(d)$.

Local distributional vanishing is equivalent to equality on compact test tuples with pairwise disjoint supports. Compactly truncate general disjoint Schwartz tuples in both L^1 and L^2, and use F.a to pass to the limit. This proves precisely the target A.

Source. RPM1, the proof of Theorem A from Theorem B in Section 2, using Lemma 2.1; final assembly in Section 9.

A.scope. Why the target retains its two qualifications

MODEL-SPECIFIC. Overview Parent A Target.

These examples are the constructions in Section 10 of RPM1. Each satisfies the stronger bound $\mathbb{E}e^{F(\chi)} \le e^{C\|\chi\|_2^2}$ and hence the exponential hypothesis of A. They are not inputs to the covariance proof.

Contact terms can fail covariance. In dimension two, use $\mathcal{T} = \mathrm{Sym}_0^2(\mathbb{R}^2)$ with orthonormal basis $(dt^2 - dx^2)/\sqrt{2}$ and $(dt \otimes dx + dx \otimes dt)/\sqrt{2}$. Give the components independent white noises with unequal variances $a, b > 0$. The characteristic functional is

$$\Phi(\chi) = e^{-\frac{1}{2}\int (a\chi_1^2 + b\chi_2^2)}.$$

The coordinate and diagonal reflections preserve this quadratic form. For one-sided tests, disjointness from the reflected supports makes each characteristic RP matrix a positive rank-one matrix. The law is stationary and Gaussian dominated, with planar weights ± 2. A $\pi/4$ rotation exchanges the two tensor components and changes their unequal variances, so the law is not rotation invariant. Its covariance defect is supported on the diagonal; nonempty moments on pairwise disjoint supports vanish.

Weight three can fail off-diagonal covariance. Let ϕ be the massive scalar Gaussian free field in $\mathbb{R}^2$ and regard $F = (\partial_t\phi, -\partial_x\phi)$ as taking values in the harmonic-cubic representation in an orthonormal basis proportional to $t^3 - 3tx^2$ and $3t^2x - x^3$. In the clockwise rotation convention, $\rho_3(R(\theta)) = R(-3\theta)$ and $\rho_3(\Theta) = \mathrm{diag}(-1, 1)$ for $\Theta(t, x) = (-t, x)$. Its square-group action intertwines with the support-preserving test map $\chi \mapsto -\operatorname{div}(\mathrm{diag}(1, -1)\chi)$. This transfers the four mirror positivities; its variance is at most $\|\chi\|_2^2$. Its weights are ± 3.

For two first-component insertions initially separated by $(r, 0)$, write $h(r)$ for the scalar covariance at $r > 0$. Their rotated two-point defect is

$$\tfrac{1}{2}(h''(r) - h'(r)/r)(1 - \cos 8\theta), \qquad h''(r) - h'(r)/r > 0.$$

Thus disjoint smooth tests detect a nonzero off-diagonal defect. Here $M_2 = 8$, compatible with the weight-three growth bound $4N$. The theorem's uniform representation cutoff cannot be enlarged to three.

Source. RPM1, Section 10. The source credits these two constructions to the supplied referee report.

A.application. The scalar lattice application as a separate use of the criterion

MODEL-SPECIFIC. Overview Parent A Target.

Conditional application. Start with a given normalized scalar sequence in Sokal's construction and a subsequential continuum probability limit ν, as specified in Section 11 of the preserved 27-page source. This conditional supplement is retained from that edition; current RPM1 omits the application. It verifies inputs for that same limit and supplies no new lattice construction.

Let

$$F_a(f) = a^d \sum_i f(ai)\varphi_i^{(a)}, \qquad G_a(k) = \mathbb{E}[\varphi_0^{(a)}\varphi_k^{(a)}] \geq 0,$$

and assume the susceptibility normalization

$$\mathsf{X}_a = \sum_k G_a(k) = \widetilde{z}\,\widetilde{m}^{-2}a^{-d}, \qquad \widetilde{z}, \widetilde{m} > 0 \text{ fixed.}$$

The source's exponential estimate and $2uv \leq u^2 + v^2$ give

$$\mathbb{E}e^{F_a(f)} \leq \exp\left(\frac{a^{2d}}{2}\sum_{i,j}|f(ai)|G_a(i-j)|f(aj)|\right) \leq \exp\left(\frac{\widetilde{z}}{2\widetilde{m}^2}a^d\sum_i|f(ai)|^2\right).$$

Exponential-moment convergence and Riemann sums pass this to

$$\mathbb{E}_\nu e^{F(f)} \leq e^{\widetilde{z}\|f\|_2^2/(2\widetilde{m}^2)}.$$

Compact truncation and Fatou extend the bound to all real Schwartz tests. This stronger pure L^2 bound implies the hypothesis of A.

Verify the remaining hypotheses on that law. Coordinate and diagonal RP hold in the same infinite-volume scalar state. The scalar embedding commutes with all origin-mirror pullbacks, and a one-sided continuum test is sampled only on that side. Along the same subsequence every finite characteristic RP matrix therefore converges entrywise to a positive matrix for ν. This same law is invariant under all real translations. Its scalar weights are zero. The target theorem applies and gives off-diagonal $O(d)$ moment covariance. Fixed positive $\widetilde{m}$ is part of the application.

Source. Historical 27-page edition, Section 11, pp. 24–25, TeX 2246–2328; the source's citations to Sokal (2.27), (2.81), (2.83), (2.86), Lemma 2.2 and the same-state RP statements.